

BE-24101

Biological Science for Engineers

About the Course



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Biological Science for Engineers

M1: Introduction to Biological Sciences

M2: Molecular Biology and Genetics

M3: Biochemistry

**M4: Applications of Biological Sciences in
Engineering**

M5: Global Challenges and Ethical Considerations

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Biological Science for Engineers

M1: Introduction to Biological Sciences

- Overview and importance of biology in engineering
- Origin of Life
- Cell Theory, Cell Structure and Function

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Biological Science for Engineers

M2: Molecular Biology and Genetics

- Central Dogma of Molecular Biology
- DNA, RNA and Protein structure and function
- Mendelian Genetics
- rDNA Technology and Genome Editing

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M3: Biochemistry

- Cell Metabolism
- Enzymes and Catalysis
- Cell Communication and Signalling

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M4: Applications of Biological Sciences in Engineering

- Biomaterials
- Bioinformatics
- Biosensors and Bioelectronics (Biological Sensors- Ear & Eye)
- Synthetic Biology
- Nanobiotechnology

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M5: Global Challenges and Ethical Considerations

- Convergence of AI and Biology
- Climate change and food security
- Biosafety and Biohazards
- Ethical Considerations

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Recommended Books

1. Lehninger A, Principals of Biochemistry
2. Stryer L, Biochemistry
3. K. Wilson & K.H. Goulding, A biologist's guide to Principles and Techniques of Practical Biochemistry
4. Biology for Engineers" by Arthur T. Johnson

Reference Books

1. Purves et al, Life: The Science of Biology
2. R. Dulbecco, The Design of Life
3. Biological Science Edited by Soper, Cambridge low price edition
4. Synthetic Biology: A Primer" by Paul S. Freemont and Richard I. Kitney
5. Introduction to Bioinformatics by Arthur Lesk
6. Genomes by T.A. Brown

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THANK YOU